FLATNESS, FINITE PRESENTATION, SMOOTHNESS AND ETALENESS OF A MAP BETWEEN PUSHOUTS DESCEND TO THE TWO INPUT MAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20016-20039 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20041-20050, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The properties-above transfer law used by the chapter's gluing-of-modifications arguments; both pushout cards are counted as a matched pair of distinct aggregates.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma pushout-along-closed-immersions (existence, card L16144); More on Algebra, lemma properties-algebras-over-fibre-product (property transfer across pushouts of rings).

Source reference. Original lemma 0B7M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19987–20004. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08KQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20050–20050. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : Z \to X$ and $j : Z \to Y$ be closed immersions of schemes. Let $f : X' \to X$ and $g : Y' \to Y$ be morphisms of schemes and let $\varphi : X' \times_{X,i} Z \to Y' \times_{Y,j} Z$ be an isomorphism of schemes over Z. Consider the morphism*

$$h : X' \amalg_{X' \times_{X,i} Z, \varphi} Y' \longrightarrow X \amalg_Z Y$$

Then we have

1. *h is locally of finite type if and only if f and g are locally of finite type,*
2. *h is flat if and only if f and g are flat,*
3. *h is flat and locally of finite presentation if and only if f and g are flat and locally of finite presentation,*
4. *h is smooth if and only if f and g are smooth,*
5. *h is étale if and only if f and g are étale, and*

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(6) *add more here as needed.*

Proof. We know that the pushouts exist by Lemma 0B7M. In particular we get the morphism h. Hence we may replace all schemes in sight by affine schemes. In this case the assertions of the lemma are equivalent to the corresponding assertions of More on Algebra, Lemma 08KQ. □